

while supported by a physical rehabilitation therapist. Conditioning also results in a decreased duration of circulating lactic acids in the body after training. Therefore the athlete may experience less or no muscle soreness when appropriately trained. Adequate rest between training intervals will decrease the concentration of lactic acid in the tissues. As a result, muscle and overuse injuries may be prevented. In addition, the application of specific treatments after training runs or during a race, such as massage, stretching, or if indicated the use of other modalities (see muscle injuries) may help to prevent injuries in the sled dog.

It is the impression of the authors that injuries are less frequently seen in teams that have covered between 3000-4000 training miles, when compared to teams that covered 1000-2000 miles during preparation for a long distance race, such as the Iditarod. It is possible that such intense training programs help to select for animals that are not very prone to injury. If a dog went through a reasonable training program over the year, and competed in several races of increasing length in the racing season without a significant injury, it is possible that it will successfully complete marathon races such as the Iditarod. However, this hypothesis needs to be proven.

Despite all preventative measures, injuries occur occasionally. It is important to detect these injuries, and to apply appropriate treatment as soon as possible.

FRACTURES

Any fracture makes it necessary to remove the dog from the race. Emergency treatment for fractures on the trail is difficult to apply. If possible, pain medication should be administered. Oral medications for pain relief and/or to reduce inflammation include tramadol, amantadine, and non-steroidal anti-inflammatories (NSAIDs), such as Rimadyl, Metacam, Etogesic etc. It is important to follow the instructions on the prescription label to prevent or decrease possible negative side effects with any drug. Trail veterinarians may have injectable analgesics such as Morphine derivatives available. If there is an open wound associated with the fracture (Figure 5), the damaged area should be treated as indicated in the chapter on wound treatment in this handbook. If this is not possible, the application of a sterile bandage, or at least clean layer of cotton should be used to cover the wound. Fractures of the humerus or femur are difficult to stabilize using a bandage. With these injuries it may be adequate to limit supportive care to the administration of analgesics and careful handling. In contrast, fractures below the elbow or stifle can usually be stabilized by application of a "Robert-Jones-Bandage" (Figure 6). This bandage consists of a sterile layer if there is a wound. Then a thick layer of soft cotton, followed by a layer of stretch gauze, and a splint are applied. A light aluminum or stable plastic rod, a straight piece of wood, or rolled newspaper could be used in addition to, or in lieu of commercially available splints. The final layer consists of vetrap, or carefully applied tape. Animals with fractures should be transported to the closest veterinary facility as soon as possible.



Figure 5: An open fracture has to be treated following the principles of open wound management.



Figure 6: Robert-Jones Bandage.

Radiographs are necessary for a definitive diagnosis (Figure 7A). Depending on various factors, definitive treatment methods include the application of a professionally applied cast or stabilization using orthopedic surgical devices, such as pins and wire, bone plates, pin-plate combinations, interlocking nails, or external fixators (Figure 7B). The prognosis for fracture healing depends on the severity of the injury, the age of the animal, the time between incident and treatment, and most importantly the post-operative care. Strict limitation of exercise for at least 8 weeks after surgery is imperative for initial bone healing to occur. With severe fractures this time may be longer. Radiographs are taken at specific time intervals to monitor bone healing. A specific activity protocol should be designed for each patient. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities. Bone plates used to stabilize fractures of bones lacking soft tissue coverage, for example the distal radius, may get very cold when the dog is outside for an extended period of time. This cold can lead to a painful sensation and lameness. To prevent or treat lameness that occurs secondary to thermal conduction, surgery to remove the implant is usually recommended. While bone healing is frequently adequate for normal leg use after approximately 8 weeks in most cases, it may take 1 year or longer to achieve adequate bone stability that allows removal of the implant. At that point, the bone should be able to withstand forces while running without the support of a bone plate.



Figure 7A:
Comminuted
femur fracture

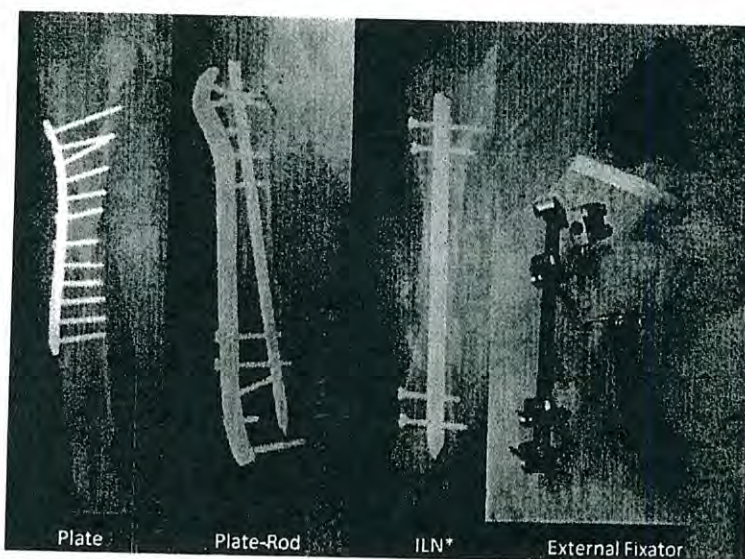


Figure 7B: Fracture repair options. *ILN=Interlocking nail.

MUSCLE INJURIES

Muscle injuries occur frequently in the sled dog. They include exertional rhabdomyolysis and acute muscle injuries such as muscle strains to certain muscle groups. Information on exertional rhabdomyolysis during endurance sled dog racing is provided elsewhere in this book. The following paragraph is written to inform the reader about possible treatment options for acute muscle injuries. The chapter on orthopedic examination provides information on how to detect such injuries. Over the last several years various techniques, used successfully in humans and horses, have been implemented in the treatment of muscle injuries in the canine athlete. However, their application may be limited during a race. The most commonly used techniques to treat muscle injuries on the trail are application of various liniments, thermal treatment, and massage. Changing harness type or fit and changing the side of the gangline the dog runs on can help reduce stress on an injured leg. More advanced therapies include range of motion exercises, exercises to improve balance and proprioception, hydrotherapy, laser therapy, neuromuscular stimulation, ultrasound, or acupuncture. Many of these modalities can also be applied during rehabilitation of other orthopedic injuries.^{1,2,5}

Liniment application

The benefit of liniment or ointment application is probably one of the most discussed topics among mushers and trail veterinarians. It is important to understand that there is no miracle drug that can be used during a race that will cure orthopedic injuries. Appropriate rest and supportive modalities are the best treatments for most soft-tissue injuries. Appropriate conditioning and careful racing strategies, especially in rough terrain and hills, are a better choice to likely

prevent injuries, rather than relying on the effect of a liniment once an injury has occurred. Various liniments are available, and it appears that each musher or veterinarian swears by one or the other. There is no scientific evidence proving one liniment superior to the other. It is imperative to verify that the ingredients of a liniment are allowed during a race prior to application of a liniment. Liniments that can be used without interfering with doping regulations contain various substances, which may or may not be beneficial to the musculoskeletal system of the sled dog. The beneficial effect of a liniment may arise solely from the additional massage that accompanies application of the liniment. Rubbing liniment on the legs, shoulders and flanks may encourage additional blood flow to the muscles, resulting in increased relaxation and elasticity. Several liniments seem to be used frequently among the sled dog community in North America. These include Algyval, Zalox, Accede, Emu oil, and others.

Algyval (Figure 8) contains mainly hyperoxygenated peanut oil and rosemary oil extract as active ingredients. The mechanism of action is not scientifically proven or understood. It has been suggested that the active ingredients inhibit leukotriene and prostaglandin receptors, resulting in an interruption of the inflammatory cascade. However, no studies are available to solidify these hypotheses. Algyval seems to have a better effect when applied warm. A possible negative side effect is that it tends to make the skin “greasy”. These areas may freeze at very low temperatures, possibly leading to skin damage from hypothermia.

Zalox (Figure 9) is also frequently used to relieve joint swelling, inflammation, and muscle soreness. Active ingredients are essential oil of wintergreen, menthol, camphor, eucalyptus oil, and tea tree oil. Other ingredients include deionized water, aloe vera, kosher glycerin, stearic acid, mineral oil, glyceryl monostearate, glycol stearate, isopropyl palmitate, dimethicone, and triethanolamine. The same precautions for preventing skin freezing as with Algyval should be taken with Zalox, especially as it has water in its base. Aloe vera may help to treat minor cuts and skin injuries.

The application of Accede or Emu oil may also have beneficial effects to treat muscle injuries, but reliable information on ingredients and their effect is scarce. Accede is oil based also, and contains several herbal extracts including rosemary. As a safety precaution, any handlers with allergies should check ingredient lists. For example, both Algyval and Accede contain peanut oil or peanut extract, which could trigger a reaction for someone with a peanut allergy. The authors know about at least one dog with confirmed allergic reaction to these products that did not react with an olive oil based liniment – *so peanut allergies appear to happen in dogs too*. That dog had redness, pruritus and alopecia in the areas it was applied.

Other liniments are available and may contain the following ingredients: willow, arnica, wormwood, devils club, and/or buckbean, grape seed, apricot kernel oil, calendula, vitamin E, or tea tree oil. It must be emphasized, that these ingredients may show beneficial effects for some dogs, but scientific data supporting the use of these substances is lacking. The use of liniments or supplements containing willow bark may cause a positive drug test for salicylates. *Always check before the race to make sure the products are allowed.*



Figure 8 (left) and Figure 9 (right): Algyval or Zalox are frequently used during massage of sore muscles. Their use may improve muscle function or prevent injury.

Thermal treatment⁵

The application of cold helps to control pain and inflammation immediately after injury. If decreased range of motion is a result of pain or swelling, apply cold. Snow or crushed ice can be placed in a plastic bag. A thin cotton cloth is placed over the injured area, and the cold pack applied for 15-20 minutes after injury. It can be reapplied after a pause of the same time. Icing the damaged tissue provides an analgesic effect, decreases blood flow, edema and muscle spasm. There is some controversy about the use of cold packs in sled dogs because the injured areas may already be hypothermic. As such, cold therapy may be reserved for surgical patients, for patients that have a very acute injury and can be treated in a warm environment, or during races in temperatures above freezing.

Heat therapy is frequently used to treat injuries in sled dogs such as strains of the shoulder muscles (see below – Figure 10). If decreased range of motion is a result of stiffness, apply heat. On the trail, custom made sleeves or wraps for the specific anatomic location are equipped with chemical handwarmers to provide heat to the injured tissues. These hot or warm packs apply superficial heat. If available in the kennel, therapeutic ultrasound or laser (see below) can be used, and may effectively provide deep heat penetration. Heat causes vasodilation which increases blood flow, resulting in increased metabolism of the treated tissues, relaxation, increased elasticity and facilitation of tissue healing. Similarly, the range of motion of treated joints may increase, resulting in more comfortable ambulation. Heat should be applied for approximately 30 minutes. However, many mushers leave packs on for longer, often the entire 6-8 hour rest period, and some even use them while running. Temperatures should not rise above 113°F (45°C) in order to prevent thermal injury.

Range-of-motion therapy⁵

Similar to stretching, range-of-motion exercises help improve muscle elasticity, decrease the risk of adhesion development between soft tissue and bone, and prevent further injury to joints, ligaments, tendons, and muscles. Range-of-motion exercises may not be as important to perform on the trail, as they are to an injured animal recovering at the kennel. Frequency and intensity of range of motion exercises depends on the specific injury and should be performed following the recommendations of a veterinarian experienced in canine sports medicine.

Massage therapy¹

Massage therapy is a technique that can be done on the trail and may result in significant improvement of the animal's performance or recovery. Several mushers have asked specifically for more information on "how to get a dog through an injury". To gain the necessary knowledge in these disciplines, we recommend that specially tailored courses be attended.^{1,2} It is not possible to perform a complete "text-book" massage on every animal at every checkpoint. However, some of the provided information may be applicable to certain animals, which will hopefully benefit from massage, and finish the race successfully. On the other hand, massage therapy could be implemented in training, possibly helping to prevent acute injuries during a race.

Benefits of massage therapy include stimulation of the lymphatic system and increased localized blood circulation, resulting in increased cellular activity and metabolism, removal of metabolic waste products, and thus decreased localized inflammation. The release of naturally occurring endorphins stimulates the parasympathetic nervous system, resulting in relaxation. A quick return of muscles to a physiologic resting length, early detection of injuries, and assisting in the promotion of healthy scar tissue formation are further benefits of massage therapy. These benefits result in improved immune function, reduced recovery time between workouts or from injuries, decreased post-workout discomfort, improved proprioceptive skills, improved work efficiency, stability and flexibility, decreased risk of stress-related injuries, and prevention of chronic conditions. Although there is no proof in the literature yet that massage can diagnose, treat, reverse, affect, or help to prognose an existing condition in the canine athlete, the benefit of massage therapy in long distance races, such as the Iditarod, seems to be undeniable. Several mushers are convinced that dogs with certain injuries, such as carpal or some shoulder problems would not have been able to finish the race without applying massage therapy at every checkpoint or even intermittently on the trail.

It is important to avoid forcing an animal to receive a massage, and never to massage directly over an open wound or skin infection, or over a suspicious lump. The animal's response must be observed, and the massage modified as needed for comfort and safety.

Every massage session consists of five stages: Opening, palpation, body of the massage, closing and stretching. Opening involves greeting, initial touching to quiet the dog, and to encourage relaxation. Palpation consists of running the hands over the entire body of the animal using mild strokes. It will help to relax the animal and provide the therapist with information on general health of the tissues. Tension, swelling, spasms, or heat may be felt. The body of the massage is the application of selected strokes and techniques, applied based on the findings from palpation. There are three categories of body massage: Maintenance of health and fitness (1-2 treatments per month), optimizing performance potential (3-4/month), or rehabilitation from injury (daily to 3-5/week). Stretching is used after the massage to restore good circulation to all parts of the body, and to help solidifying the work performed. Closing is performed using light strokes to reconnect

all aspects of the body, and to provide a calm and relaxed feeling. Ending the massage too abruptly may lead to an unresolved or confused conscious state in the animal.

There are six strokes that are all part of the body of the massage. They can all be modified in speed and pressure to alter the effect that is desired for each individual patient. Faster strokes are more stimulating, slower strokes more sedating. The therapist will have to strive for excellence by practicing, practicing, and practicing again. Results will improve with time and experience.

Passive touch is an excellent opening stroke. It is sustained contact, using the hand, fingers or fingertips. It can be used to feel for tissue irritation, it can help to create good contact between therapist and animal, and has a calming effect. It can be applied with only one hand, while the other hand is applying another stroke elsewhere. Passive touch is applied with minimal pressure and increases neural awareness and the sensation for warmth.

Effleurage describes passing of the hand over tissues with moderate pressure. It warms an area prior to advanced work and can be used for opening or closing, and transition between areas. It stimulates lymphatic flow and superficial circulation. If massage therapy is performed with focus on one area, effleurage is a good technique to provide the remainder of the body with a few comforting strokes.

Petrissage is also known as kneading. It describes the application of deep pressure to the tissues or lifting of tissues, leading to increased circulation. This helps with removal of waste products. It should only be applied after passive touch and effleurage have been performed. If petrissage is too aggressive, it may lead to the damage of tissues. Always begin with gentle pressure or lifting and a slow rhythm. One-handed petrissage is the lifting and squeezing of tissue into the palm of one hand. The other hand can be used to apply a different stroke, or to restrain the animal. Small muscle groups or the limbs may be massaged using this stroke. Two-handed petrissage describes the placement of a muscle belly between the palms of the therapist's hands. The muscle is lifted and squeezed carefully. The amount of applied force has to be monitored carefully. Open-C / Closed-C petrissage describes a two-handed petrissage in which the thumbs and forefingers are held in the shape of a C. Tissues are lifted and pressed together with one toward the other hand, resulting in the creation of an effective fluid pump within the muscle belly. This technique is useful to relieve a mild cramp. Several more variations of petrissage are available.

Compression describes the application of pressure with a flat or balled hand, the arm, elbow or fingertips of the therapist to the underlying tissues. It is a useful stroke for muscles that are difficult to lift. Pressure intensity should be increased carefully from light to deep. Depending on the body part used to apply pressure, the effect will differ: A flat hand provides pressure over a large area, whereas the elbow creates a more focused pressure point. Application of compression using the fingertips over a joint may help to stimulate fluid activity and relax surrounding tissues.

Friction strokes increase warmth and localized circulation, relax restricted muscles and lengthen muscle fibers. This stroke needs to be applied with caution as not to damage healthy tissue. It should only be used after opening strokes such as passive touch or effleurage. Direct pressure can be applied over areas of tenderness, adhesions, or as part of professional acupressure. Repetitions with increasing pressure can be applied. However, this stroke should be limited to five applications for not more than 30 seconds. If no improvement is observed, the area should be left, and repeated treatment applied at another time. Overly aggressive treatment may result in tissue damage. Several sub-categories of friction are described: Rubbing is used to build up heat in the superficial tissues. Chucking is used on the limbs and consists of holding tissues with the whole hand, and carefully pulling the tissues proximally and distally. Twisting is performed by the same hand placement, but the tissues are carefully twisted. These techniques also increase localized circulation. Circular friction is used to warm and mobilize underlying soft tissues, and supplies a relaxing effect on the nervous system. Linear friction helps to move waste products out of muscle bundles. The stroke follows the direction of the muscle fibers. Cross-fiber friction describes pressure application across the direction of muscle fibers. It is used to loosen adhesions or hardened muscle areas. Direct pressure for a few seconds drains blood from the area. A short release of pressure is then followed by movement of the fingers across the muscle fibers for a few seconds. These steps are repeated several times with the goal being to resolve the adhesion or "knot". Cross-fascicular friction is the spanning of muscle fiber bundles using the fingertips. This technique leads to a more diffuse distribution of pressure and is frequently used to treat adhesions, which cannot be released using other techniques.

Tapotement is the application of short percussive strokes. Indications for tapotement are stimulation of weak or atrophied tissue, numbing of tissues, engaging the attention of the animal or loosening and warming tissues up prior to work. It can be applied by drumming the fingers on certain areas (tapping) or by applying short strokes with the flat hand onto the abdominal or thoracic muscles (slapping). To warm tissues up prior to exercise, the edge of the hand can be used to carefully hit the underlying tissues (hacking). To possibly relieve congestion in the chest or to decrease joint swelling, the hand can be formed to a small cup and light blows are applied to the area of interest (cupping). Finally, the fingers can be moved over the tissues in a "chopstick" – like fashion. This stimulates superficial circulation and lymphatic drainage, helping with removal of waste products. Contraindications for tapotement are muscle spasms, cramps or hypertonicity. The application of this technique to such conditions may worsen the severity of the problem.

Finally, the limb, skin, or certain body parts can be loosened by shaking it. "Chucking" describes moving a fascial plane up and down. "Vibrating" is the application of short pulses from the palm or fist of the therapist to joints or deeper tissues. The pulse waves penetrate the tissues and lead to a sensation of warmth. They also stimulate joint fluid production, which may be beneficial prior to work. "Brushing" is a short stroke in which the whole the hand is used to touch the skin rapidly, with the effect of restoring sensation and increasing circulation. These techniques are good for closing the massage. After each massage, the optimal effect can be reached by stretching the tissues for five to ten minutes and a period of easy walking.

Stretching results in increased muscle length, and thus flexibility. Increased flexibility results in a decreased amount of injuries of both, muscles and joints. Therefore, good stretching is imperative for injury prevention. The following guidelines should be followed when stretching: *Never stretch a cold muscle*. Some light work or massage will increase blood flow and decrease muscle fiber rupture from sudden application of stretching to cold tissues. *Always stay within a comfortable range of motion*. Passive repetitive stretches consist of fast movement of muscle through its complete range of motion. This stretch does not involve excessive tension, which is held for an extended period of time, as is done with static stretching. Passive repetitive stretching creates a fluid pump within the muscle, leading to removal of waste products and supplying oxygen and nutrients to the muscles. Animals seem to be more receptive to this stretching technique than to static stretching. From neck to hind legs, several stretches are available for the major muscle groups, and can be applied safely on a regular basis to improve flexibility. If possible, stretching should be performed 3-5 times per week. Neck stretches consist of careful lateral flexion, extension, and dorsal/ventral flexion. Front leg stretches consist of flexion/extension, abduction/adduction, internal/external rotation of the shoulder, flexion/extension of the elbow, wrist, and digits. Stretches for the spine and abdomen consist of careful dorsal/ventral flexion and extension as well as lateral flexion. Hind leg stretches consist of flexion, extension, abduction, adduction, internal, external rotation of the hip and extension/flexion of stifle, tarsus, and digits.

Laser Therapy (LT)⁵

LT uses electromagnetic radiation that enhances energy production (Adenosine-Tri-Phosphate, ATP) and metabolism within cells that may ultimately result in an increase of events that accelerate tissue healing (increased amount of growth factors, cytokines, and cell replication). It also seems to provide analgesia, to preserve cartilage properties, and to help resolve peripheral nerve injuries. The U.S. Food and Drug Administration (FDA) has approved 635nm low-level lasers for the management of chronic minor pain, such as osteoarthritis and muscle spasm. The use of laser type (helium-neon or gallium-aluminum-arsenide with 632.8nm to 780-904 nm, respectively), and applied energy ($4-11 \text{ J/cm}^2$) vary, depending on the injury. LT can be used daily during week 1 after injury, and every other day during the second week. The use of LT remains controversial, but is noninvasive and has not been associated with side effects, when used properly. Higher doses of laser energy may inhibit tissue healing. Therefore, this treatment should only be applied by specifically trained personnel.

Neuromuscular electrical stimulation (NMES) and Transcutaneous electrical nerve stimulation (TENS)⁵

NMES is used to stimulate skeletal muscle contractions through excitation of nerves. Differently pulsed patterns are used to treat various orthopedic and occasionally spinal cord injuries. The application of NMES results in increased blood circulation, muscle strength, joint mobility, and decreased tissue contracture, edema, and pain.⁵ TENS is the use of electrical stimulation for pain control. The effectiveness of these treatments has been proven for dogs. Specific protocols for strengthening or pain control need to be followed to avoid injury. These treatments are only to be applied by specifically trained personnel.

Therapeutic ultrasound (TUS)⁵

The application of energy through an ultrasonic beam leads to compression and separation of the exposed tissues. As a result, deep tissue heating occurs, which has been shown to result in tissue healing. TUS may be of benefit during the rehabilitation of muscle injuries, joint contractures, or during the healing process of bone, cartilage, or tendons, when applied properly. The intensity, frequency, and duration of ultrasonic waves can be adjusted, depending on the treated injury. Frequencies from 0.25-MHz to 3-MHz with an intensity of $0.5-2.0 \text{ W/cm}^2$ are frequently used. The recommended transducer speed movement is 4cm/s. Treatment is usually applied for 10 days daily, then on an as needed basis. It is recommended not to exceed two 10-day courses without a 3-week rest. Additional therapies such as massage and stretching aid to achieve the optimal effect of TUS. This treatment is only to be applied by specifically trained personnel.

Extracorporeal shock wave treatment (ESWT)⁵

ESWT is the application of acoustic waves initiated outside the body. The clinical effects include decreased swelling and inflammation, realignment of tendon fibers, analgesia, increased bone and soft tissue healing through increased

neovascularization and vascularity. Appropriate use is imperative to prevent negative side effects. ESWT has been used successfully in horses for treatment of tendinopathies, back pain, suspensory ligament desmitis, osteoarthritis, and stress fractures. In dogs, ESWT has been applied successfully to treat shoulder tendinopathies with tissue calcification, and osteoarthritis of hips and elbows. Both, subjective and objective evaluation showed improvement of the underlying condition after ESWT was applied.⁵ The effect was compared to the effect that is typically expected after the use of NSAIDs. Although promising, further studies need to be conducted to support this treatment modality for muscle injuries in the canine athlete. This treatment is only to be applied by specifically trained personnel.

Acupuncture

Acupuncture may improve healing of muscle injuries or other orthopedic conditions in some selected patients.⁶ The musculoskeletal system is like a pulley and lever mechanism - a system of agonists and antagonists. Orthopedic injury may result in a conflict between the opposing agonist and antagonist muscle groups, which creates undue stress on the joints, muscles, tendons and ligaments. Obscure muscular pain found in association with either acute traumatic or chronic/overuse injuries can result in weakness at the injured site, and to the onset of lameness after a certain degree of activity. In traditional Chinese medicine, acupuncture points are utilized to address these complaints. Acupuncture points, important for the treatment of muscle injuries, are trigger points and motor points. A trigger point is defined as a "hyperirritable spot in the skeletal muscle that triggers a characteristic referred pain or tenderness phenomenon." Some of the indicators of a trigger point include pain referral, muscle twitch response and tight palpable bands. These tight and frequently painful palpable bands may be treated effectively with the acupuncture needle. A motor point is defined as "the most electrically excitable area of the muscle, and represents the greatest concentration of nerve endings", medically speaking the neuromuscular junction. The insertion of an acupuncture needle into a motor point may elicit a change in length of the muscle, resulting in muscle relaxation. Acupuncture to the motor points seems to correct abnormal muscle function and reflexive spasm. The motor point serves as a way of balancing the agonist and antagonist muscle groups. Acupuncture performed by an experienced therapist may help to decrease discomfort, and possibly treat muscle injuries in the canine athlete. However, there is no scientific data available to support this treatment method, and skepticism in its effectiveness may be warranted.

Hydrotherapy

Although this treatment method can only be applied if an appropriate beach, river, pool or rehabilitation center for canine athletes are available, its benefits justify a short description in this chapter. It is becoming more common for mushers to use a pool to exercise their dogs during the off season to maintain conditioning. Water temperature can influence the cardiovascular system. It is generally recommended to keep the water temperature during hydrotherapy for dogs in the range of 78.8° to 82.4°F (26° to 28° C). Studies to determine the optimal water temperature for sled dogs need to be conducted. The effect of gravity is reduced in direct relation to the level of submersion. Body weight decreases by 9%, 15%, or 62% if the water level rises to tarsus, stifle, or greater trochanter, respectively. This effect is described as buoyancy. It describes the force lifting the body upward while submerged. With increased buoyancy, there is less impact on the musculoskeletal system when walking. This prevents injury to the joints or injured tissue. The amount of hydrostatic pressure, which increases with submersion, may improve lymphatic or venous drainage from injured tissues or joints. The viscosity of water, the depth of immersion, and turbulent drag resists limb and body movement. This may result in muscle strengthening and improved cardiovascular function. It has been shown that with movement just beneath the surface of the water, approximately eight times the amount of energy are required to double forward motion. Some specifically designed hydrotherapy treadmills offer the use of jets, which can be used to increase the aforementioned effects. Thus, muscle workout can be increased in a very controlled manner, and re-injury prevented by appropriate application. In summary, hydrotherapy assists weightbearing, increases range of motion, balance, proprioception, strength, endurance, and cardiovascular fitness, and decreases swelling of tissues or joints.

Exercises to improve balance and proprioception⁵

Several exercises are available to improve balance and conscious proprioception (see above, conditioning). These exercises not only help to prevent injuries, but can also be applied during the recovery period from an injury. Most of these techniques cannot be performed during a race. Their application is limited to use at the kennel. Explaining the specifics of these techniques is beyond the extend of this chapter. However, the interested reader is encouraged to increase his knowledge through readily available literature, or by participating in specially designed courses or classes (see above).

SHOULDER INJURIES

The diagnosis of shoulder injuries is made by an appropriate orthopedic examination (see previous chapter).